

DUSHYANTA DHYANI

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Senior Applied Scientist with 10+ years of experience in deploying ML solutions for business applications. Currently technical lead for Amazon Pricing's Foundational modeling team for a billion-scale catalog. Research spans information retrieval, LLM-based probabilistic regression, efficient LLM inference, and agentic systems for automated auditing. Expert in building and leading cross-functional teams.

EXPERIENCE

Senior Applied Scientist, *Amazon Pricing*

Sep 2024 - Present

Technical lead for a cross-functional team, re-imagining Amazon's product price estimation strategy.

- Developed a retrieval-augmented system for price distribution estimation using similar-product pricing signals at billion-scale across marketplaces and locales.
- Launched an LLM-based price quality assessment system for channel-agnostic pricing
- Developed a hyper-parallel LLM attribute extraction pipeline achieving 90%+ inference cost reduction
- Guided development of a centralized MLOps hosting and deployment platform
- Conceptualized and implemented Evaluation Driven Science, a framework for grounding roadmap prioritization in systematic model evaluation

Senior Applied Scientist, *Amazon Books*

April 2024 - Aug 2024

Applied Scientist II

Oct 2020 - March 2024

Applied Scientist I

Oct 2019 - Sept 2020

- **Inauthentic Book Duplicates** - Led a cross-functional team to drive the science solution and long-term strategy for the Inauthentic Book Duplicates program across Amazon Books. Includes E2E design and implementation of a multi-stage text ranking service:
 - Transformer-based neural ranking model for book pair scoring
 - Scalable hybrid retrieval engine with lexical and vector search.
 - Near-real-time inference service scoring 500M book pairs per day.
- **Book Content Moderation** - Built the Content Moderation framework for detecting hate speech and explicit content in long-context book interiors using multi-task, multi-instance learning (MIL) approaches.

Machine Learning Engineer, *Alexa Brain & Amazon Comprehend*

July 2018 - Sep 2019

Contributed to NLP model development, inference infrastructure, and annotation workflows across Alexa's domain classification and Amazon Comprehend's relation extraction and custom classification services.

SELECTED PUBLICATIONS

Theodore Glavas, Nikhita Vedula, **Dushyanta Dhyani**, Antonios Valkanias, Yilun Zhu, Shervin Malmasi, "Intra-Prompt Parallel Decoding for Common-Context Question Answering", *EMNLP 2026*.

Nikhita Vedula, **Dushyanta Dhyani**, Bryan Wang, Shervin Malmasi, "Scaling E-Commerce Attribute Extraction with Parallel Decoding", *11th Workshop on Automated Knowledge Base Construction (AKBC) 2026*.

Yilun Zhu, Yuan Zhuang, Nikhita Vedula, **Dushyanta Dhyani**, Shaoyuan Xu, Moyan Li, Mohsen Bayati, Bryan Wang, Shervin Malmasi, "Text-to-Distribution Prediction with Quantile Tokens and Neighbor Context", *ACL 2026*.

Theodore Glavas, Nikhita Vedula, **Dushyanta Dhyani**, Yilun Zhu, Shervin Malmasi, "Breaking the Autoregressive Chain: Hyper-Parallel Decoding for Efficient LLM-Based Attribute Value Extraction", *Findings of ACL 2026*.

Nikhita Vedula*, **Dushyanta Dhyani***, Laleh Jalali, Boris N. Oreshkin, Mohsen Bayati, Shervin Malmasi, "Quantile regression with large language models for price prediction", *Findings of ACL 2025*.

TECHNICAL SKILLS

ML/AI Frameworks: PyTorch, PyTorch Lightning, HuggingFace Transformers, numpy, scikit-learn

LLM & NLP: LLM fine-tuning, SFT, RAG, information retrieval, probabilistic regression, efficient inference

Large-Scale ML Systems: PySpark, Airflow, AWS SageMaker with distributed training and inference, AWS Trainium

EDUCATION

M.S. CSE (Thesis Track) : *The Ohio State University*

Aug 2016 - May 2018

Thesis: Boosting Supervised Neural Relation Extraction with Distant Supervision (*Adviser*: Prof. Huan Sun
Committee Member: Prof. Alan Ritter)